

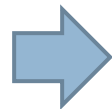


PRICE PROMOTIONS & CUSTOMER LIFETIME VALUE

Overview

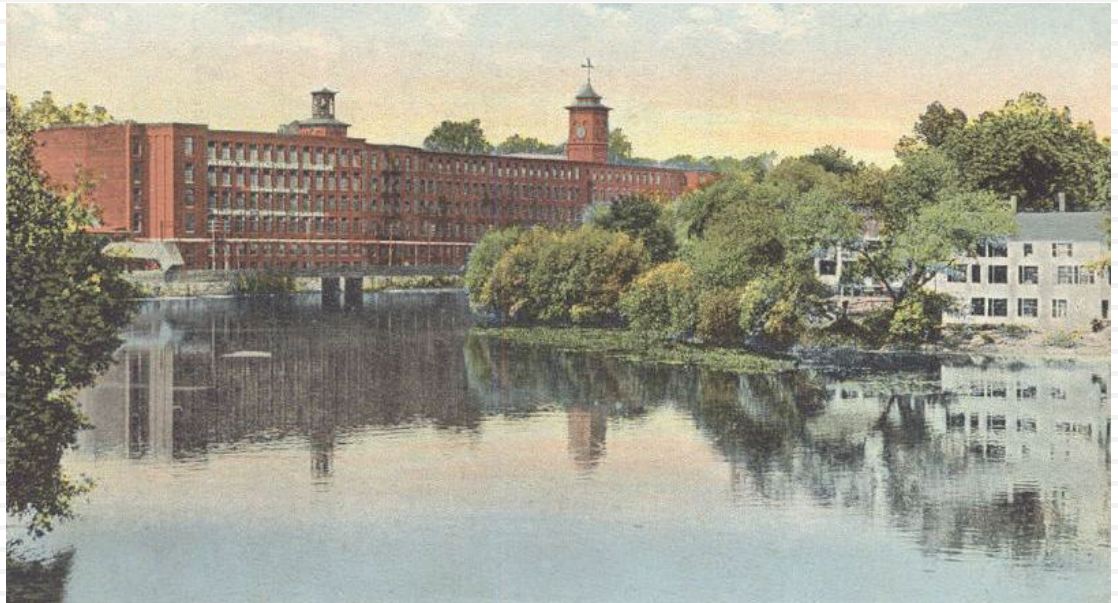
2

- Price Promotions
 - ▣ One-shot segmentation
 - versus
 - ▣ Long-run investment
- Pricing as an **investment**
 - ▣ Pricing objectives & the **lifetime value of the customer**
 - ▣ Pricing decisions to balance **acquisition & retention**



Pricing & CRM

3 Case Study: Nashua Photo



See Appendix for examples of coupons as **segmented pricing**

Evaluating Price Promotions

Example: Nashua Photo

4

Evaluating Price Promotions

Example: Nashua Photo

4

- Based on 1993 *Efficiency Report*, how does Nashua evaluate its price promotion campaigns?

Evaluating Price Promotions

Example: Nashua Photo

4

- Based on 1993 *Efficiency Report*, how does Nashua evaluate its price promotion campaigns?

Promotion Number	Mailer Type	Circ	CPM	Total Resp.	New Cust.	Total Cost	Cost per Resp.
Agfa-Showtime	FREE	336	\$0.00	14196	6678	\$3,286	\$0.23
AMA free Dev Offer	FREE	5	\$0.00	211	99	\$49	\$0.23
Corpus Chirsti Caller	59	81	\$20.78	3044	852	\$2,475	\$0.81
Cambden Courier-Post	59	318	\$12.36	8030	2248	\$7,041	\$0.88
Action Arena	TO	6	\$38.50	315	123	\$290	\$0.92
The Times Herald	59	72	\$34.96	3157	990	\$3,221	\$1.02
Coachman Deluxe	59	10	\$0.00	93	52	\$98	\$1.05
Parents	59/89	315	\$16.50	7518	4042	\$8,278	\$1.10
Freedom Press	59	40	\$19.64	1017	399	\$1,177	\$1.16
Advantage Promotions	59/89	435	\$16.50	9861	5301	\$11,432	\$1.16
Bonnie Walton	89	6	\$0.00	47	22	\$59	\$1.25
Baby's first foto	59/89	132	\$37.40	4972	2784	\$6,228	\$1.25
Lansing Journal	59	20	\$16.50	401	189	\$526	\$1.31
Pittstown Press	59	72	\$21.98	1702	705	\$2,287	\$1.34
Wheeling Reader	59	390	\$34.50	12498	3079	\$17,268	\$1.38
Christian Books	59	100	\$24.20	2418	1083	\$3,398	\$1.41
Winn Dixie	59	1624	\$28.05	42752	20110	\$61,436	\$1.44

Average cost per respondent across campaigns is \$2.40



[Click to see entire list of campaigns](#)

Evaluating Price Promotions

Example: Nashua Photo

5

- How effective are these campaigns?
 - ▣ Avg promotional GM = \$1.07

Evaluating Price Promotions

Example: Nashua Photo

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Return on
marketing
investment

Promotion Number	Mailer Type	Circ	CPM	Total Resp.	New Cust.	Total Cost	Cost per Resp.	Gross Margin	Total Return	ROMI
Agfa-Showtime	FREE	336	\$0.00	14196	6678	\$3,286	\$0.23	-\$1.75	-\$24,843	-8.56
AMA free Dev Offer	FREE	5	\$0.00	211	99	\$49	\$0.23	-\$1.75	-\$369	-8.55
Corpus Chirsti Caller	59	81	\$20.78	3044	852	\$2,475	\$0.81	\$0.97	\$2,953	0.19
Cambden Courier-Post	59	318	\$12.36	8030	2248	\$7,041	\$0.88	\$0.97	\$7,789	0.11
Action Arena	TO	6	\$38.50	315	123	\$290	\$0.92	\$1.97	\$621	1.14
The Times Herald	59	72	\$34.96	3157	990	\$3,221	\$1.02	\$0.97	\$3,062	-0.05
Coachman Deluxe	59	10	\$0.00	93	52	\$98	\$1.05	\$0.97	\$90	-0.08
Parents	59/89	315	\$16.50	7518	4042	\$8,278	\$1.10	\$1.01	\$7,585	-0.08
Freedom Press	59	40	\$19.64	1017	399	\$1,177	\$1.16	\$0.97	\$986	-0.16
Advantage Promotions	59/89	435	\$16.50	9861	5301	\$11,432	\$1.16	\$1.01	\$9,962	-0.13
Bonnie Walton	89	6	\$0.00	47	22	\$59	\$1.25	\$1.32	\$62	0.06
Baby's first foto	59/89	132	\$37.40	4972	2784	\$6,228	\$1.25	\$1.09	\$5,403	-0.13
Lansing Journal	59	20	\$16.50	401	189	\$526	\$1.31	\$0.97	\$389	-0.26
Pittstown Press	59	72	\$21.98	1702	705	\$2,287	\$1.34	\$0.97	\$1,651	-0.28
Wheeling Reader	59	390	\$34.50	12498	3079	\$17,268	\$1.38	\$0.97	\$12,123	-0.30
Christian Books	59	100	\$24.20	2418	1083	\$3,398	\$1.41	\$0.97	\$2,345	-0.31
Winn Dixie	59	1624	\$28.05	42752	20110	\$61,436	\$1.44	\$0.97	\$41,469	-0.32



Click to see entire list of campaigns

Evaluating Price Promotions

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Action Arena	TO	6	\$38.50	315	123	\$290	\$0.92	\$1.97	\$621	1.14
The Times Herald	59	72	\$34.96	3157	990	\$3,221	\$1.02	\$0.97	\$3,062	-0.05
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Parents	59/89	315	\$16.50	7518	4042	\$8,278	\$1.10	\$1.01	\$7,585	-0.08
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Click to see entire list of campaigns

Only 4 campaigns with ROMI>0

Evaluating Price Promotions

Example: Nashua Photo

6

- What about cannibalization of bounce-back mailers?
Suppose 10% of *repeats* would have used bounce-back...

Evaluating Price Promotions

Example: Nashua Photo

Many users are likely not new customers, resulting in cannibalization.

and

6

- What about **cannibalization** of bounce-back mailers?
Suppose 10% of *repeats* would have used bounce-back...

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AMA free Dev Offer	FREE	5	\$0.00	211	99	\$49	\$0.23	-\$1.75	-\$395	-9.07
Corpus Chirsti Caller	59	81	\$20.78	3044	852	\$2,475	\$0.81	\$0.97	\$2,455	-0.01
Cambden Courier-Post	59	318	\$12.36	8030	2248	\$7,041	\$0.88	\$0.97	\$6,477	-0.08
Action Arena	TO	6	\$38.50	315	123	\$290	\$0.92	\$1.97	\$577	0.99
The Times Herald	59	72	\$34.96	3157	990	\$3,221	\$1.02	\$0.97	\$2,570	-0.20
Coachman Deluxe	59	10	\$0.00	93	52	\$98	\$1.05	\$0.97	\$81	-0.17
Parents	59/89	315	\$16.50	7518	4042	\$8,278	\$1.10	\$1.01	\$6,796	-0.18
Freedom Press	59	40	\$19.64	1017	399	\$1,177	\$1.16	\$0.97	\$846	-0.28
Advantage Promotions	59/89	435	\$16.50	9861	5301	\$11,432	\$1.16	\$1.01	\$8,927	-0.22
Bonnie Walton	89	6	\$0.00	47	22	\$59	\$1.25	\$1.32	\$56	-0.04
Baby's first foto	59/89	132	\$37.40	4972	2784	\$6,228	\$1.25	\$1.09	\$4,906	-0.21
Lansing Journal	59	20	\$16.50	401	189	\$526	\$1.31	\$0.97	\$341	-0.35
Pittstown Press	59	72	\$21.98	1702	705	\$2,287	\$1.34	\$0.97	\$1,425	-0.38
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Christian Books	59	100	\$24.20	2418	1083	\$3,398	\$1.41	\$0.97	\$2,042	-0.40
Winn Dixie	59	1624	\$28.05	42752	20110	\$61,436	\$1.44	\$0.97	\$36,330	-0.41



Click to see entire list of campaigns

Evaluating Price Promotions

Example: Nashua Photo

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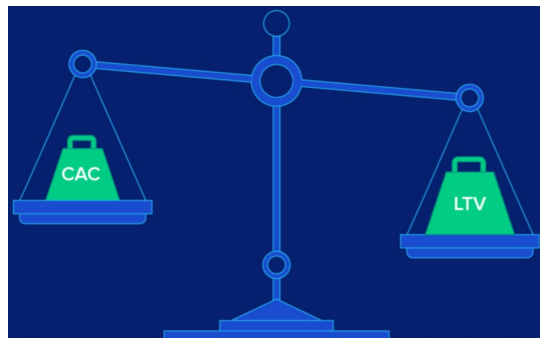
Only 1 campaigns with ROMI>0

Long Run Effects of Promotions

7

- Price promotion induces **trial**
- Trial may lead to future **repeat** purchases

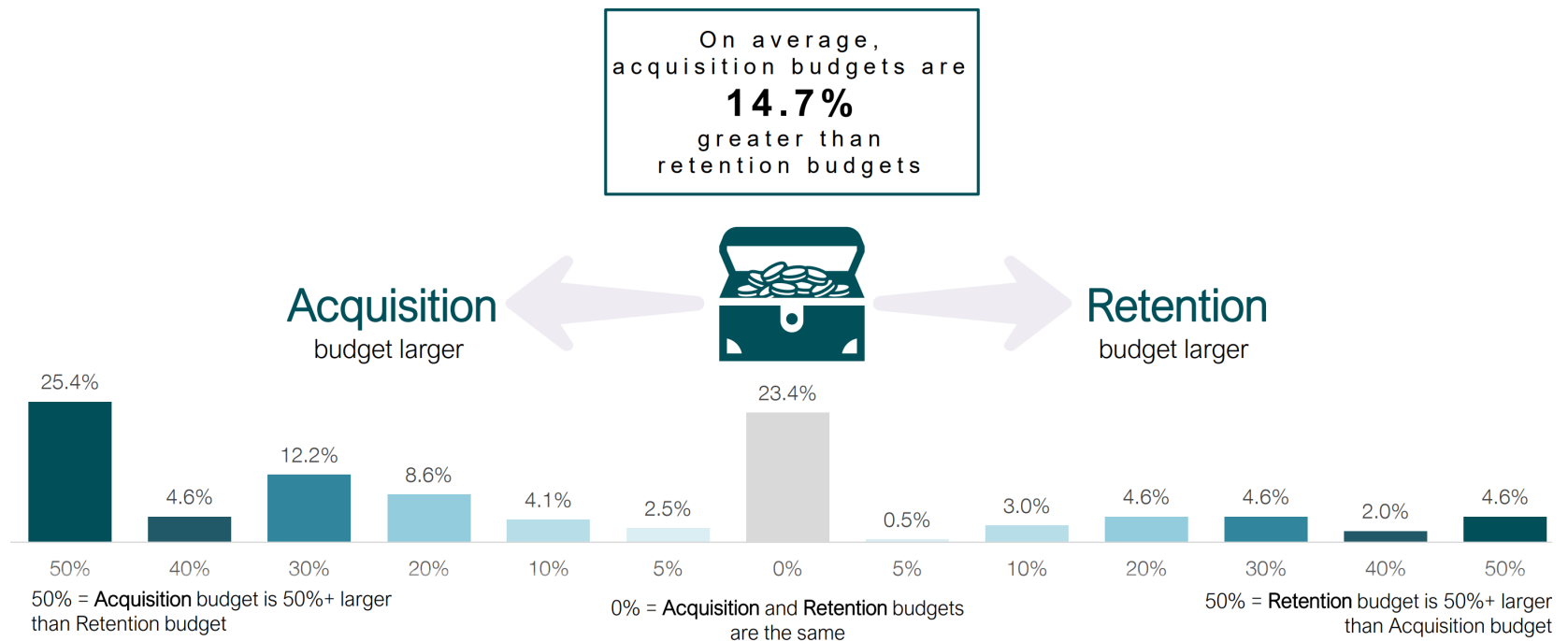
*How much should firm be willing to **discount** price to acquire a new customer?*



“Companies spend significantly more of their budgets on customer acquisition than on customer retention”

8

How do your company's budgets for customer acquisition and customer retention compare?



Source: [CMOSurvey.org](https://cmosurvey.org), 2022

Customer Lifetime Value (CLV)

Simple NPV Calculation for customer's *cash-flow*

9

What is CLV of a customer who will spend \$10 each period (starting today) over some time horizon (e.g., annuity)?

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Simple NPV Calculation for customer's *cash-flow*

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Infinite horizon (i.e., perpetuity)

Customer Lifetime Value (CLV)

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What is CLV of a customer who will spend \$10 each period (starting today) over some time horizon (e.g., annuity)?

Infinite horizon (i.e., perpetuity)

$$NPV = \$10 + \left(\frac{1}{1+r}\right) \$10 + \left(\frac{1}{1+r}\right)^2 \$10 + \dots$$

r = interest rate
i.e., Firm's cost of capital

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discount factor $\longrightarrow \delta = \frac{1}{1+r}$

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$$NPV = \$10 + \delta \$10 + \delta^2 \$10 + \dots$$

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$$\begin{aligned} NPV &= \$10 + \delta \$10 + \delta^2 \$10 + \dots \\ &= \frac{\$10}{1-\delta} \end{aligned}$$

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$$NPV = \$10 + \delta \$10 + \delta^2 \$10 + \dots$$

$$= \frac{\$10}{1-\delta}$$

$$\Rightarrow NPV = \$10 + \delta NPV$$

this recursive formulation
will be useful shortly.

LR promotion benefit: **Customer Journey**

example: Nashua Photo

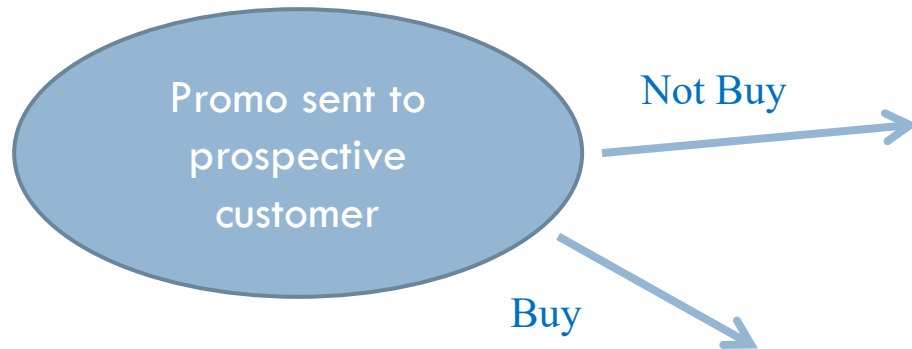
10

Promo sent to
prospective
customer

LR promotion benefit: **Customer Journey**

example: Nashua Photo

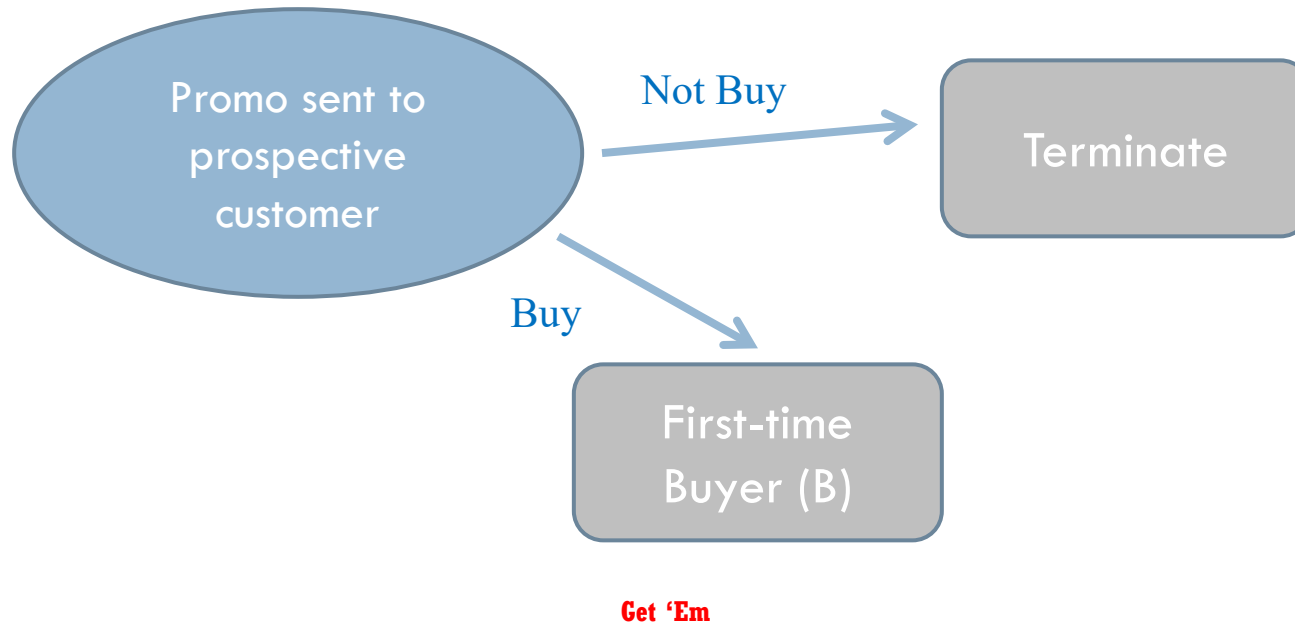
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LR promotion benefit: **Customer Journey**

example: Nashua Photo

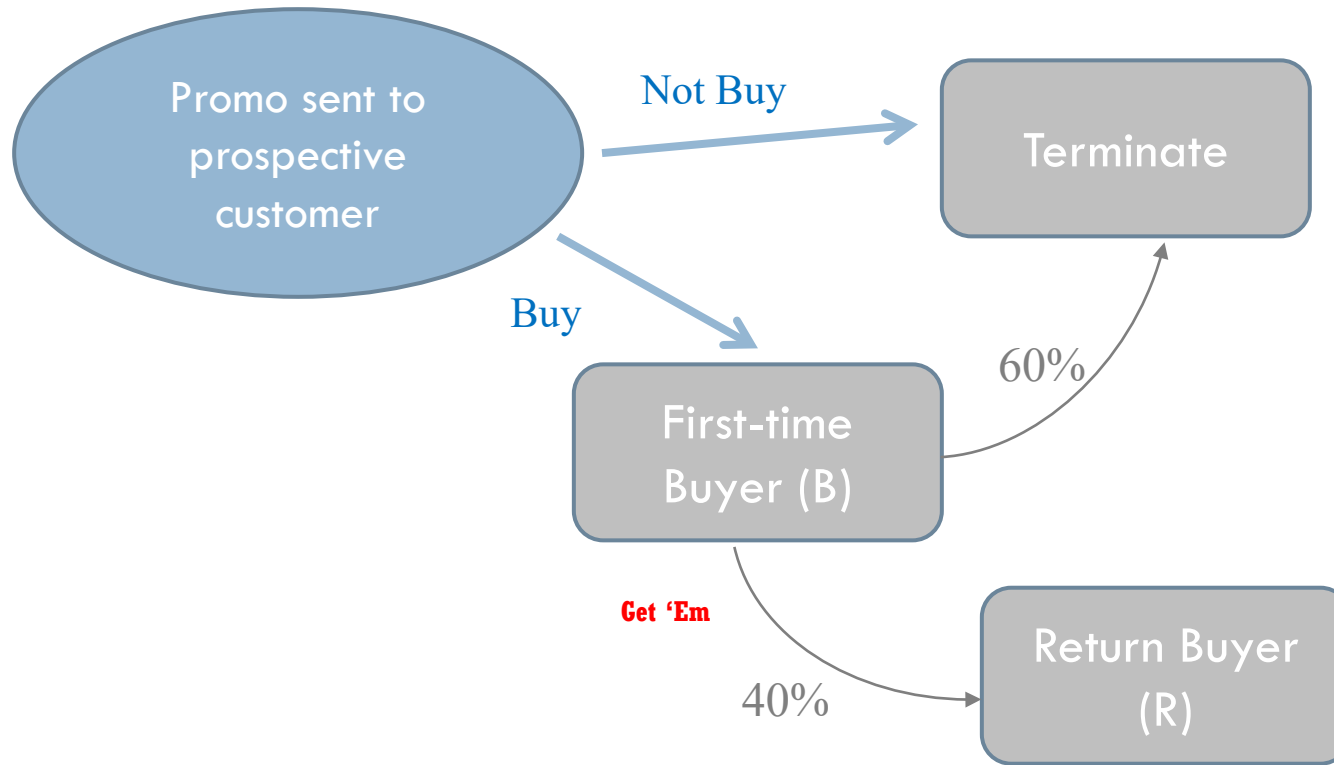
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LR promotion benefit: **Customer Journey**

example: Nashua Photo

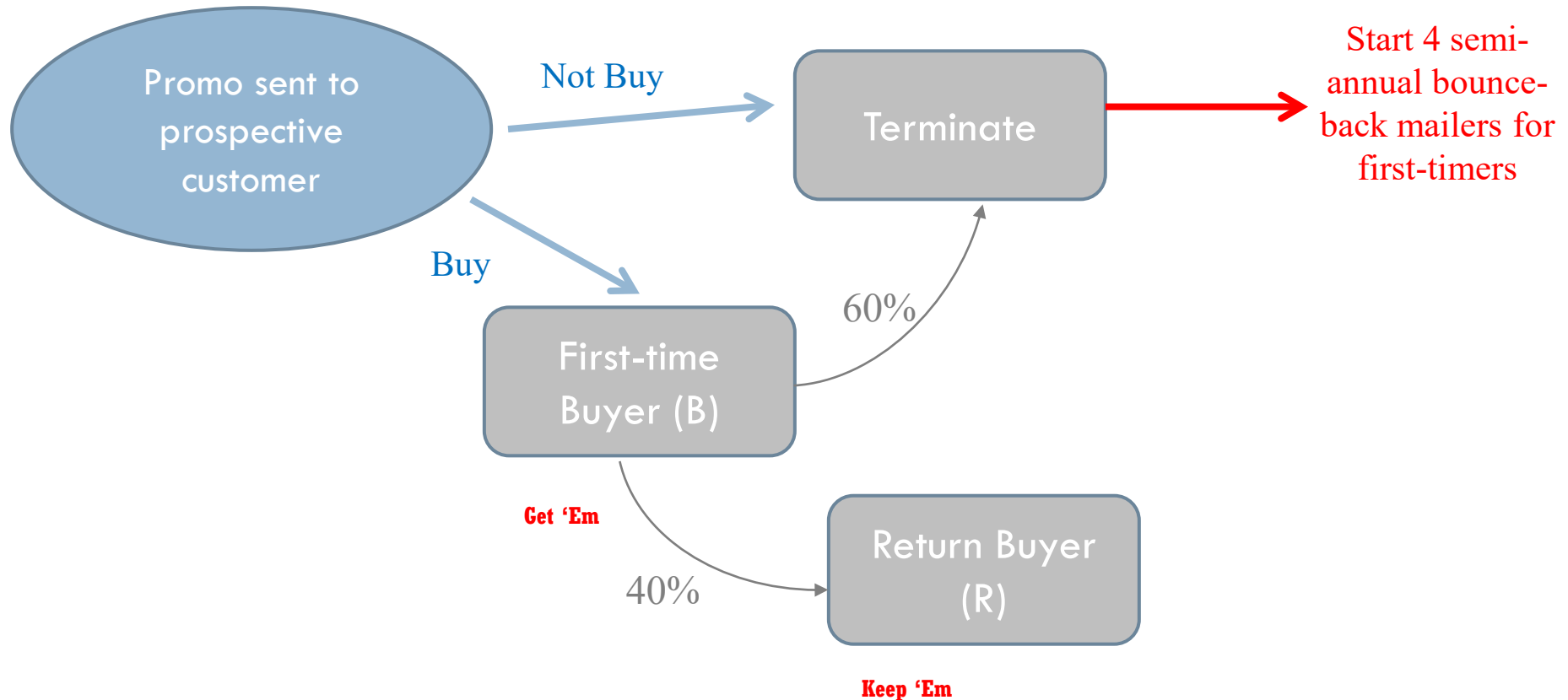
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LR promotion benefit: **Customer Journey**

example: Nashua Photo

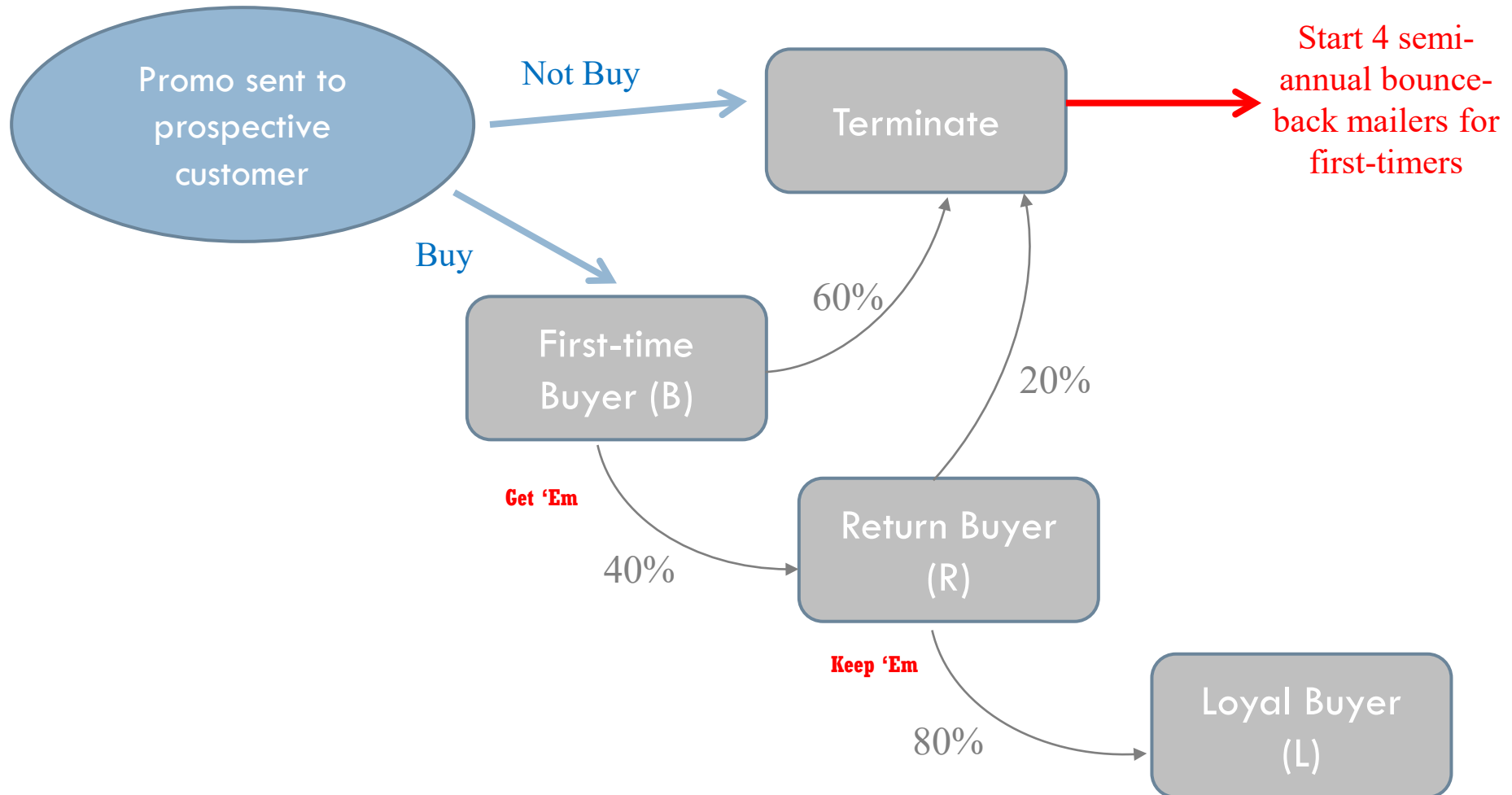
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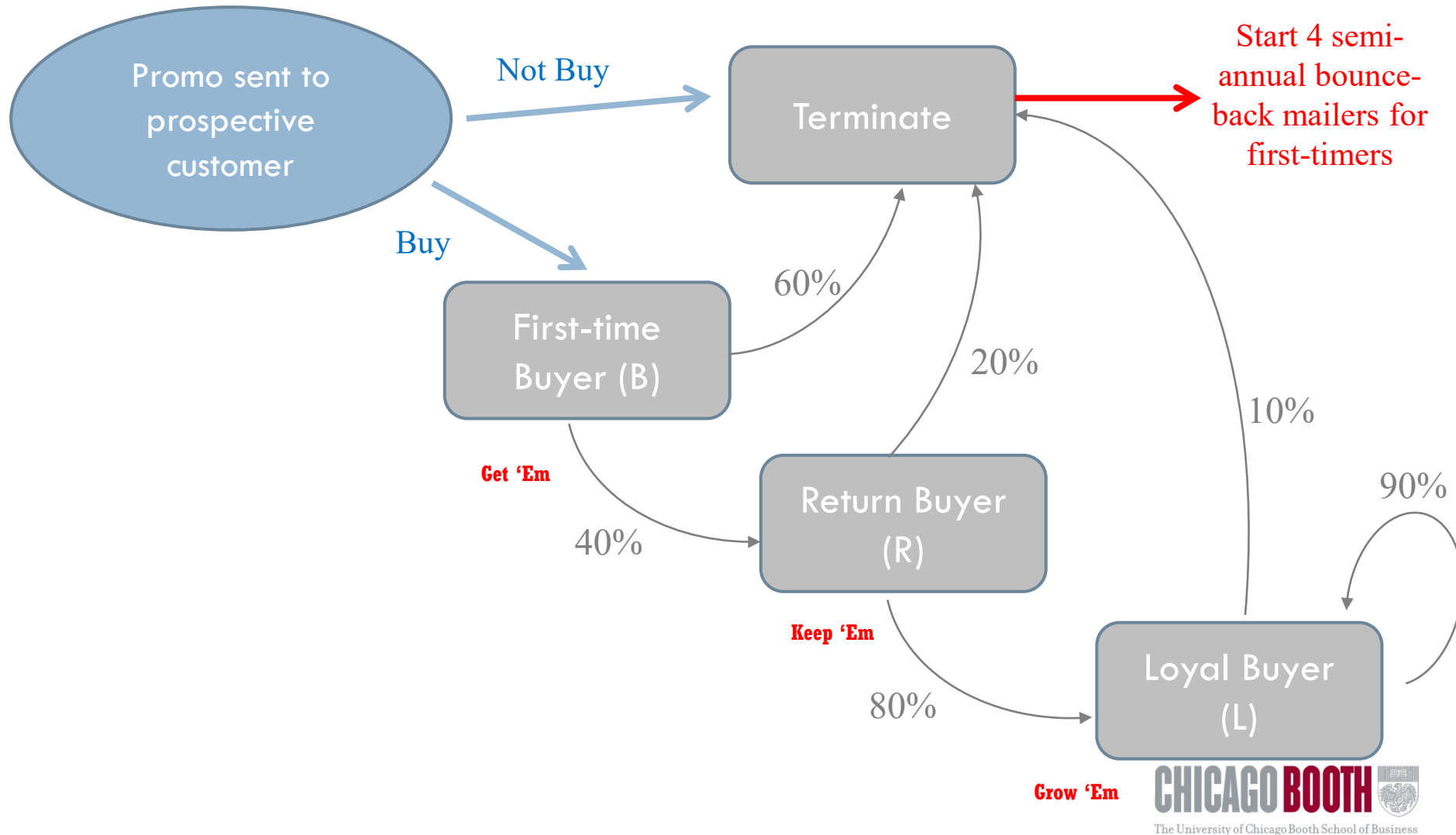
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LR promotion benefit: **Customer Journey**

example: Nashua Photo

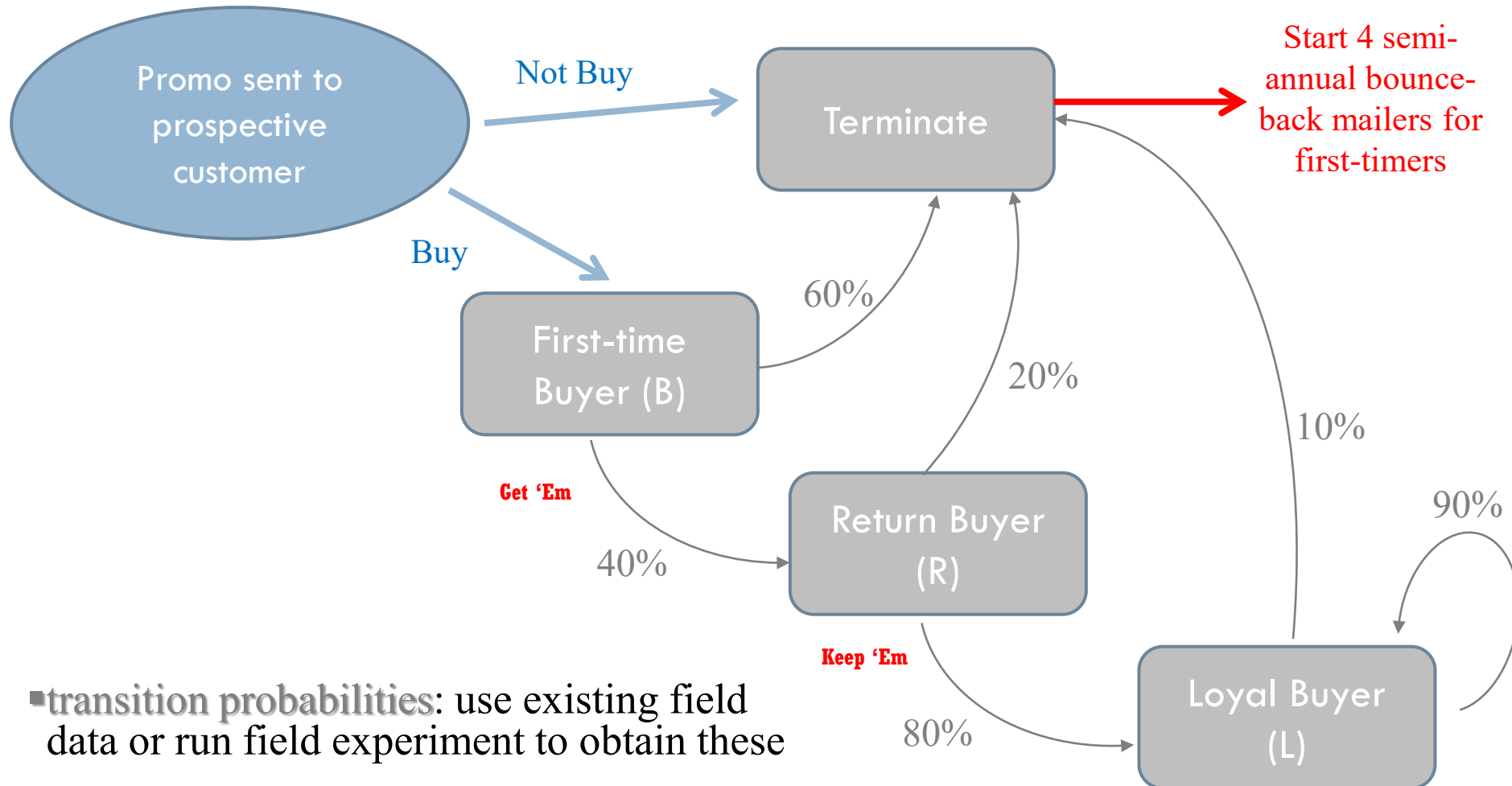
10



LR promotion benefit: **Customer Journey**

example: Nashua Photo

10



- **transition probabilities:** use existing field data or run field experiment to obtain these

Grow 'Em

Assumptions to measure LR benefits

example: Nashua Photo

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- Customer purchase cycle: 6 months
- Discounting
 - ▣ Semi-annual discount rate: $r = 4.88\%$
 - ▣ Semi-annual discount factor:

$$\delta = \frac{1}{1+r} = 0.95$$

- Bounce-back mailers (BB):
 - ▣ Cost per bounce-back mailer: \$0.35
 - ▣ PV of mail cost:

$$c = 0.35 + \delta 0.35 + \delta^2 0.35 + \delta^3 0.35 = \$1.31$$

- ▣ Assume can't convert churned customers back
- Gross Margin per repeat order:

$$m = \$2.27$$

CLV of a Buyer model

example: Nashua Photo's 3 buyer types

12

First-time
buyer

$$\text{➤ } B = \text{Prob}(\text{not Buy}|B)(-c) + \text{Prob}(\text{Buy}|B)(m + \delta R)$$

Repeat
buyer

$$\text{➤ } R = \text{Prob}(\text{not Buy}|R)(0) + \text{Prob}(\text{Buy}|R)(m + \delta L)$$

Loyal
buyer

$$\text{➤ } L = \text{Prob}(\text{not Buy}|L)(0) + \text{Prob}(\text{Buy}|L)(m + \delta L)$$

where

$$c = \$1.31$$

$$m = \$2.27$$

$$\delta = 0.95$$

$$\text{Prob}(\text{Buy}|B) = 0.4$$

$$\text{Prob}(\text{Buy}|R) = 0.8$$

$$\text{Prob}(\text{Buy}|L) = 0.9$$

Need to know the value of customer at every stage, then we can backward calculate the value of customer at the start of the relation.

CLV of a Buyer model

example: Nashua Photo's 3 buyer types

13

First-time buyer ➤ $B = 0.6 \times (-\$1.31) + 0.4 \times (\$2.27 + 0.95R)$

Repeat buyer ➤ $R = 0.2 \times (\$0) + 0.8 \times (\$2.27 + 0.95L)$

Loyal buyer ➤ $L = 0.1 \times (\$0) + 0.9 \times (\$2.27 + 0.95L)$

➡ 3 equations & 3 unknowns: B , R and L

☐ *Can you solve it?* (you could literally use pencil & paper)

☐ *Can you interpret the solution?*

CLV of a Buyer

example: Nashua Photo's 3 buyer types

14

		New State			
		<i>Terminate</i>	<i>New</i>	<i>Return</i>	<i>Loyal</i>
Current State	<i>Terminate</i>	1.00	-	-	-
	<i>New</i>	0.60	-	0.40	-
	<i>Return</i>	0.20	-	-	0.80
	<i>Loyal</i>	0.10	-	-	0.90

Cost of BB mailers	\$	(1.31)
AGM on Order	\$	2.07
Discount Rate		4.88%
Discount Factor		0.95

Lifetime Value		
1st-time Buyer (B)	\$	4.49
Return (R)	\$	11.67
Loyal (L)	\$	13.13

Click to see CLV analysis via solver (steps in Appendix)

CLV of a Buyer

example: Nashua Photo's 3 buyer types

14

		New State			
		<i>Terminate</i>	<i>New</i>	<i>Return</i>	<i>Loyal</i>
Current State	<i>Terminate</i>	1.00	-	-	-
	<i>New</i>	0.60	-	0.40	-
	<i>Return</i>	0.20	-	-	0.80
	<i>Loyal</i>	0.10	-	-	0.90

Cost of BB mailers	\$	(1.31)
AGM on Order	\$	2.07
Discount Rate		4.88%
Discount Factor		0.95

Lifetime Value		
1st-time Buyer (B)	\$	4.49
Return (R)	\$	11.67
Loyal (L)	\$	13.13

Click to see CLV analysis via solver (steps in Appendix)

➡ *get'em* promotion is investment in acquisition of 1st-time Buyers (B) where $CLV^B = \$4.49$

Evaluating Price Promotions with CLV

Example: Nashua Photo

15

- Which coupons are valuable when positive CLV is added?

Evaluating Price Promotions with CLV

Example: Nashua Photo

15

□ Which coupons are valuable when positive CLV is added?

Promotion Number	Mailer Type	Circ	CPM	Total Resp.	New Cust.	Total Cost	Cost per Resp.	Gross Margin	Total Return	ROMI
Agfa-Showtime	FREE	336	\$0.00	14196	6678	\$3,286	\$0.23	-\$1.75	\$3,459	0.05
AMA free Dev Offer	FREE	5	\$0.00	211	99	\$49	\$0.23	-\$1.75	\$50	0.03
Corpus Chirsti Caller	59	81	\$20.78	3044	852	\$2,475	\$0.81	\$0.97	\$6,284	1.54
Cambden Courier-Post	59	318	\$12.36	8030	2248	\$7,041	\$0.88	\$0.97	\$16,578	1.35
Action Arena	TO	6	\$38.50	315	123	\$290	\$0.92	\$1.97	\$1,130	2.90
The Times Herald	59	72	\$34.96	3157	990	\$3,221	\$1.02	\$0.97	\$7,019	1.18
Coachman Deluxe	59	10	\$0.00	93	52	\$98	\$1.05	\$0.97	\$315	2.22
Parents	59/89	315	\$16.50	7518	4042	\$8,278	\$1.10	\$1.01	\$24,959	2.02
Freedom Press	59	40	\$19.64	1017	399	\$1,177	\$1.16	\$0.97	\$2,639	1.24
Advantage Promotions	59/89	435	\$16.50	9861	5301	\$11,432	\$1.16	\$1.01	\$32,747	1.86
Bonnie Walton	89	6	\$0.00	47	22	\$59	\$1.25	\$1.32	\$155	1.65
Baby's first foto	59/89	132	\$37.40	4972	2784	\$6,228	\$1.25	\$1.09	\$17,416	1.80
Lansing Journal	59	20	\$16.50	401	189	\$526	\$1.31	\$0.97	\$1,190	1.26
Pittstown Press	59	72	\$21.98	1702	705	\$2,287	\$1.34	\$0.97	\$4,593	1.01
Wheeling Reader	59	390	\$34.50	12498	3079	\$17,268	\$1.38	\$0.97	\$23,821	0.38
Christian Books	59	100	\$24.20	2418	1083	\$3,398	\$1.41	\$0.97	\$6,909	1.03
Winn Dixie	59	1624	\$28.05	42752	20110	\$61,436	\$1.44	\$0.97	\$126,696	1.06

Evaluating Price Promotions with CLV

Example: Nashua Photo

15

□ Which coupons are valuable when positive CLV is added?

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Accounting for CLV, 103 campaigns with ROMI>0!!!
(over 80% of campaigns!)

Price Promotion & CLV of customer

Example: Free Unlimited Service at Sprint

16

“Sprint’s Latest Salvo in Wireless Price War: Free, Unlimited Service”

Ryan Knutsen, *Wall Street Journal*, 6-13-2017

- One year free service if switch from rival
 - ▣ Unlimited text, data, calling

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 - ▣ Offer only advertised online and sign-up online
 - ▣ Lower acquisition cost than other forms of advertising

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- Sprint’s argument:
 - ▣ Offer only advertised online and sign-up online
 - ▣ Lower acquisition cost than other forms of advertising
- Concern:
 - ▣ Targeted at Verizon customers
 - ▣ competitive reaction!

Free?

CLV Pricing and **Marketing for Bad**

Example: Free OxyContin & **addiction**

17

- Purdue used a “patient starter coupon program”
 - ▣ Free limited-time subscription for 7 to 30 days
 - ▣ 34,000 coupons redeemed nationally in 2001



OXYCONTIN
COUPON

Pharmacy Instructions:

Submit as a primary claim (cannot be processed as secondary) using the pharmacy processing information. For processing questions and comments please call the Pharmacy Helpline below.

Processing Info:

Member ID: 5IKLTW2QWU

RxBIN: 610709

RxGRP: RXCPNR4

Drug Coupon
NEVER EXPIRES

THIS IS NOT INSURANCE

Customer Service: 877-321-6755
Pharmacy Helpline: 800-223-2146





CLV Pricing and **Marketing for Bad**

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CLV Pricing and **Marketing for Good**?

Example: Netflix and inattention

18

“Netflix’s Plan to Auto-Cancel Subscriptions Is Radically Sane”

Brian Barrett, *Wired*, May 24, 2020

“If you don’t watch anything on Netflix for a year after you join, the company will send you an email asking if you want to keep your membership. You’ll also get a message if you go two years without watching anything on Netflix at any point. If you write back to say, “Yes, please keep charging me for all these seasons of Riverdale I’m ignoring,” nothing changes. If you ask to cancel or don’t write back at all, Netflix automatically stops charging you.”

The Netflix logo, consisting of the word "NETFLIX" in red, bold, sans-serif capital letters, centered within a black rectangular background.

CLV Pricing and **Marketing for Good**?

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The Netflix logo, consisting of the word "NETFLIX" in red, bold, sans-serif capital letters, centered within a black rectangular background.

CLV Pricing: acquisition & retention

Example: Ziprecruiter.com

47

- Recall online recruiting B2B pricing experiment
- Originally charging \$99 per month
- August 2015, tested 10 randomized price levels
- Determined \$281 **optimal** in short-run

Can \$99 be rationalized under CLV?

Pricing and CLV at Ziprecruiter

Pricing & Retention

20

- What about customer **retention**?
 - ▣ Cash flows from all subsequent months customer survives
 - ▣ Plausible that retention decreases for higher prices
- So retention is not a fixed number, it's a **function** of price

Pricing and CLV at Ziprecruiter

Measuring Retention

21

- Immediately after experiment, don't observe retention rates
- So need an assumption
- For instance, assume at price p :

$$Prob(trial|p) = Prob(retention|p)$$

▣ Monthly discount rate: $r = 0.4\%$ (i.e., annual rate = 5%)

➡ Monthly discount factor: $\delta = \frac{1}{1+r} = 0.996$

See Appendix for retroactive analysis using actual retention rates

Pricing and CLV at Ziprecruiter

Assume retention rate same as acquisition rate

22

- Expected Net Present Value of revenues at a given price p :

Pricing and CLV at Ziprecruiter

Assume retention rate same as acquisition rate

22

- Expected Net Present Value of revenues at a given price p :

$$NPV = p \text{Prob}(\text{buy}|p) + \delta p [\text{Prob}(\text{buy}|p)]^2 + \delta^2 p [\text{Prob}(\text{buy}|p)]^3 + \dots$$

Pricing and CLV at Ziprecruiter

Assume retention rate same as acquisition rate

22

- Expected Net Present Value of revenues at a given price $\mathbf{p}$:

$$NPV = \mathbf{p}Prob(buy|\mathbf{p}) + \underbrace{\delta \mathbf{p}[Prob(buy|\mathbf{p})]^2}_{\text{survives 2 months}} + \underbrace{\delta^2 \mathbf{p}[Prob(buy|\mathbf{p})]^3}_{\text{survives 3 months}} + \dots$$

Probability customer survives for t months:

$$Prob(buy|\mathbf{p}) \times \dots \times Prob(buy|\mathbf{p}) = [Prob(buy|\mathbf{p})]^t$$

Pricing and CLV at Ziprecruiter

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Pricing and CLV at Ziprecruiter

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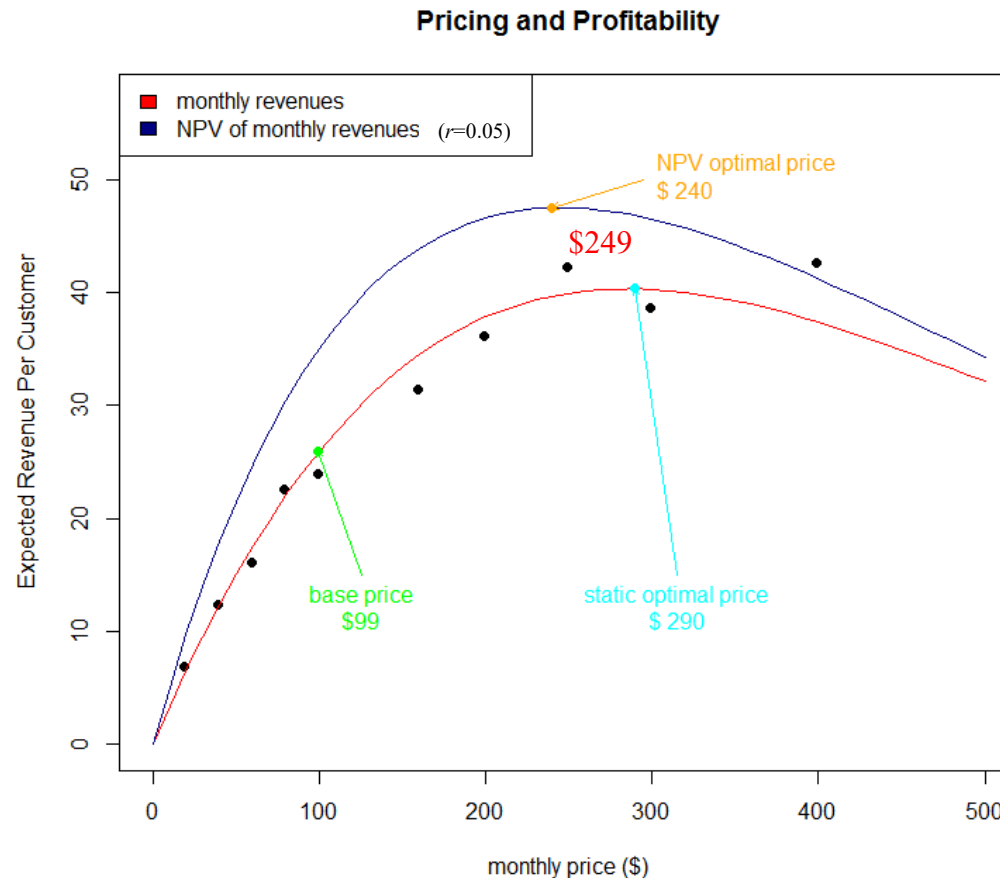
Probability customer survives for t months:
 $\text{Prob}(\text{buy}|\mathbf{p}) \times \dots \times \text{Prob}(\text{buy}|\mathbf{p}) = [\text{Prob}(\text{buy}|\mathbf{p})]^t$

Recall: $\text{Prob}(\text{buy}|\mathbf{p})$ can be estimated via logistic regression
(see module 2 and Appendix)

Pricing and CLV at Ziprecruiter

Assume retention rate same as acquisition rate

23



See Appendix for R code

Optimal lifetime value price slightly lower: \$240

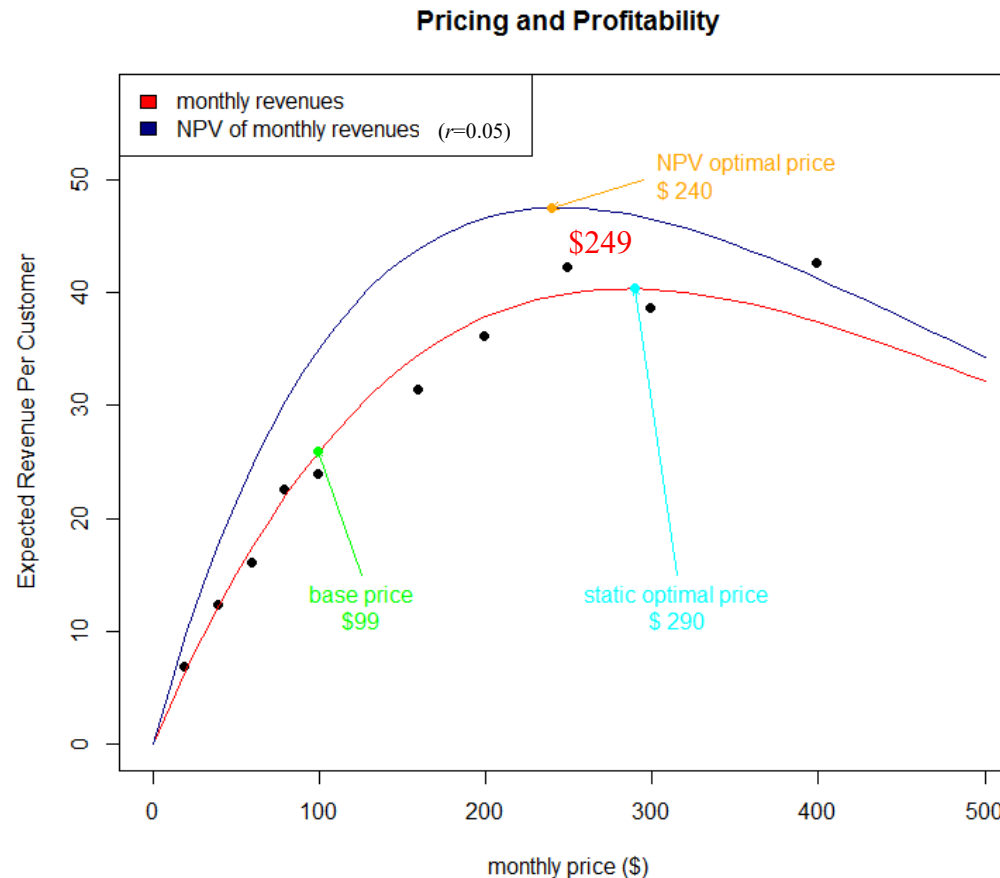
Pricing and CLV at Ziprecruiter

Assume retention rate same as acquisition rate

23

Note:
Pricing to CLV did not improve Ziprecruiter's profits much due to relatively low repeat rates

Growth mainly from static optimization as in weeks 2 & 5



Will 99 give us longer relation with paying customers?

optimal price falls from 249 slightly to 240.

static vs dynamic (require excel Goal Seek).

See Appendix for R code

Optimal lifetime value price slightly lower: \$240

Pricing and CLV at Amazon

24

Example of pricing with the future in mind



The 20 Smartest Things Jeff Bezos Has Ever Said

Words from the wise.

Morgan Housel ([cmfhousel](#))
Sep 9, 2013 at 5:09PM



10. “We’ve done price elasticity studies, and the answer is always that we should raise prices. WE don’t do that, because we believe – and we have to take this as an article of faith – that by keeping our prices very, very low, we earn trust with customers over time, and that that actually does maximize free cash flow over the long term.”

Pricing and CLV at Tesla?



Penetration pricing or price/turf war?

25

“Tesla Earnings Dented by Price Cuts”

Several rounds of reductions in the first months of the year weighed on Tesla’s first-quarter results

Rebecca Elliott, WSJ, April 19, 2023



“We’ve taken a view that pushing for higher volumes and a larger fleet is the right choice here versus a lower volume and higher margin...”

“We're the only ones making cars that technically could sell for zero profits now and yield tremendous profits in future through autonomy [and charging and software]...”

Elon Musk, CEO Tesla

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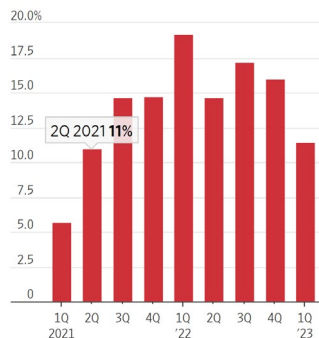


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Elon Musk, CEO Tesla

Tesla operating margin



Note: Q1 2023 is an estimate
Source: the company, FactSet



Tesla Is Pulling Back From EV Charging, and People Are Freaking Out

By [Jennifer Hiller](#), WSJ, May 1, 2024

Take-Away

26

- Discounts: *short-run gain* vs **CLV** (i.e., investment)
- Pricing to the **customer Lifetime Value (CLV)**
- Pricing and **CRM** – balancing **acquisition & retention**

Razor and blades; First price at low cost, and we get back over time (retention).

Think *Razors & Blades* over time (initial trial = Razor)

- See Appendix for “*Freemium*” as pricing to CLV

27

Extra

Coupons can also be used for static Segmented Pricing (like a product line)

Promotions as a “One Time Gain”

Example: Coupons...*But is it always profitable?*

28

- 2 types of customers who vary in disutility of coupon

<u>Segment</u>	<u>WTP</u>	<u>Disutility</u>	<u>Size</u>
H	\$15	-\$5	100
L	\$11	-\$4	100

- What's the optimal uniform price?
 - ▣ Strategy 1: Shelf Price=\$15, profit: \$1500
 - ▣ Strategy 2: Shelf Price=\$11, profit: \$2200
- What's the optimal coupon price?
 - ▣ hint: This is just product line pricing...

Promotions as a “One Time Gain”

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 - ▣ Strategy 3:
 - Coupon face value: \$4.99
 - Shelf Price: $\$7 + \$4.99 = \$11.99$
 - Profit = $\$1199 + \$700 = \$1899$

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- Profit = $\$1199 + \$700 = \$1899$

Promotions as a “One Time Gain”

Exercise

29

- Would you ever offer both a coupon and a price discount at the same time?
- 2 types of customers who vary in disutility of coupon

<u>Segment</u>	<u>WTP</u>	<u>Disutility</u>	<u>Size</u>
H	\$15	-\$5	100
L	\$7	-\$1	100

30

Extra

Using *Solver* in Excel to compute CLV of consumers at different stages of relationship

The solver add-in

31

In Excel:

1. Go to “Options,” then “add-ins” and select “solver”
2. Under “Manage” select “Excel Add-ins”
3. In menu above, select “Solver Add-in”
4. Click “Go”
5. Make sure “Solver Add-in” is checked and hit “ok”

The spreadsheet

32

Transition Probabilities

How to read transition probabilities:

For a New customer, there is a 40% chance they become a returning customer.

For a Loyal customer, there is a 10% chance they Terminate

You can adjust the transition probabilities

Solving for the LTV

From each state, you can calculate a LTV.

Use Solver to do this.

You can adjust the transition probabilities, cost of mailing, AGM on Order, Discount Rate (anything in orange).

		New State			
		Terminate	New	Return	Loyal
Current State	Terminate	1.00	-	-	-
	New	0.60	-	0.40	-
	Return	0.20	-	-	0.80
	Loyal	0.10	-	-	0.90

Cost of BB mailers	\$	(1.31)	← You can vary these.
AGM on Order	\$	2.07	
Discount Rate		4.88%	
Discount Factor		0.95	

Lifetime Value

First-time Buyer (B)	\$	-
Return (R)	\$	-
Loyal (L)	\$	-

Lifetime Value Equations

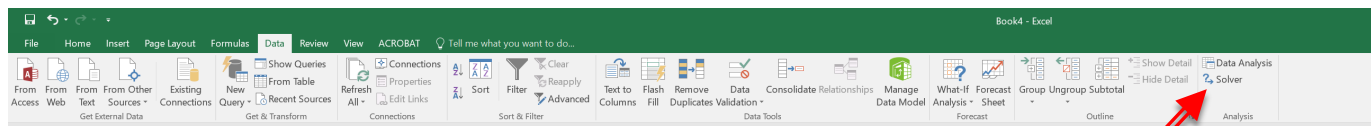
	Equation	Solver
New	\$ 0.04	\$ 0.04
Return	\$ 1.66	\$ 1.66
Loyal	\$ 1.86	\$ 1.86

- This is the spreadsheet built into the slides

Use solver to compute CLV of consumer types B, R and L

33

1. Go to “Data” tab and click on “Solver” in “Analysis” group



The screenshot shows the Microsoft Excel interface with the 'Data' tab selected. The 'Analysis' group in the ribbon contains the 'Solver' button, which is highlighted by a red arrow. The spreadsheet content includes instructions on how to read transition probabilities, a table of transition probabilities, a table of costs, and a table of lifetime values.

Transition Probabilities

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AGM on Order	\$	2.07
Discount Rate		4.88%
Discount Factor		0.95

You can vary these.

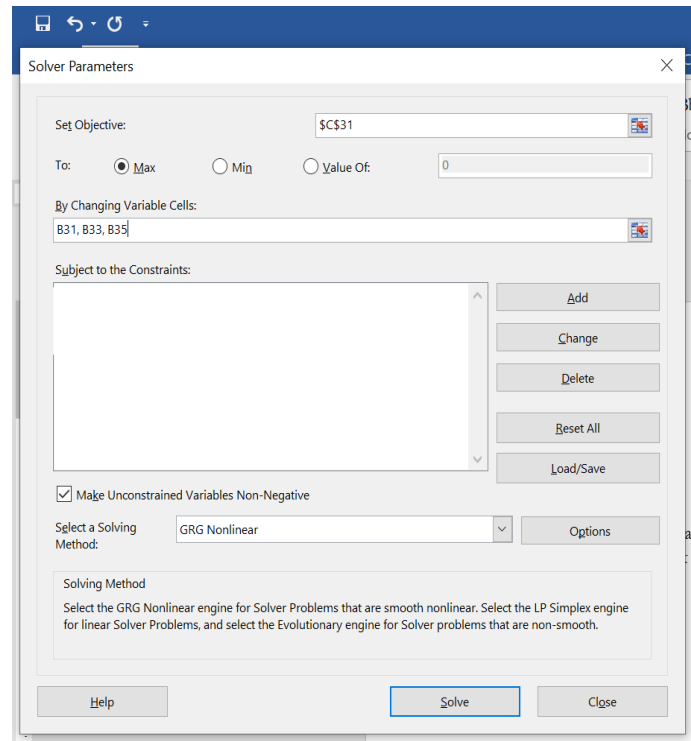
Lifetime Value	
First-time Buyer (B)	\$ -
Return (R)	\$ -
Loyal (L)	\$ -

Lifetime Value Equations		
	Equation	Solver
New	\$ 0.04	\$ 0.04
Return	\$ 1.66	\$ 1.66
Loyal	\$ 1.86	\$ 1.86

Enter equations into solver

34

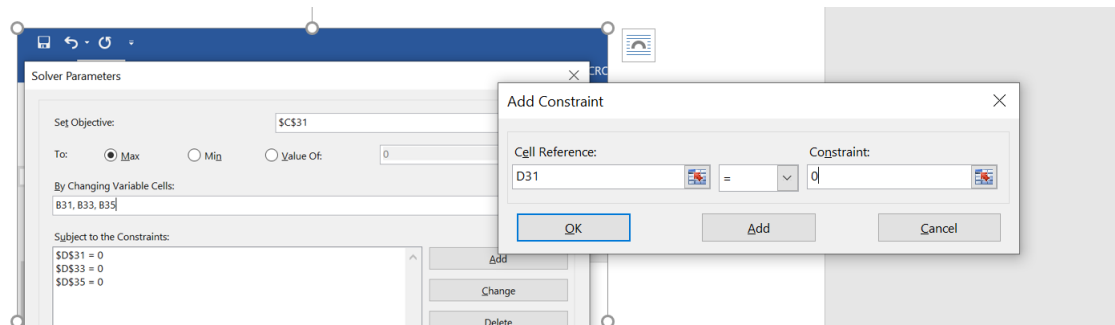
2. In “Set Objective” box, insert cell reference C31, where we set formula for first equation of system.
3. Select the objective as “Max”.
4. Insert range of cell references B31, B33 and B35 in the “By Changing Variable Cells” box.



Enter constraints

35

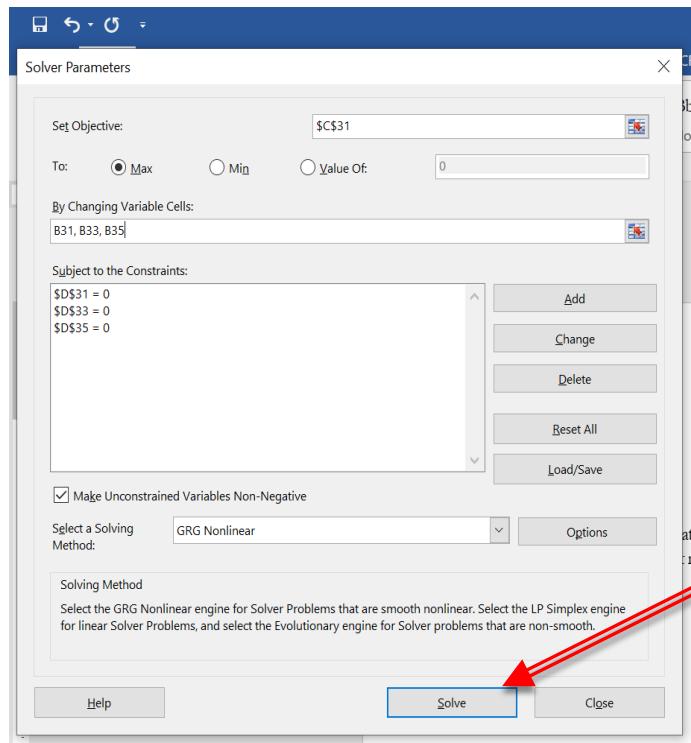
5. In “Subject to Constraints” box, add constraints for each of the equations by clicking on “Add” button.
 - ▣ In “Add Constraints” window, enter the constraints for all equations one by one by pressing “Add” button. Press OK button when done with adding all the constraints.



Now solve the system

36

6. Press “Solve” button



38

Freemium as Pricing and LTV

Freemium and LTV

39

- Many software firms offer **freemium** versions to new customers
 - ▣ Free to induce trial
 - ▣ Then charge for upgrade as user's needs grow

Is Freemium A Business Model?

The freemium is usually a growth and branding strategy rather than a business model. A free service is provided to a majority of users, while a small percentage of those users convert into paying customers through either marketing or sales funnel. The free users not converting in customers help spread the brand.



Let's formalize this model ...

Freemium and LTV

40

- freemium model:

- F = free version, with ad revenues $\$a$

- P = paid version, with price $\$p$

} Usually $a < p$

- Assume: after upgrading to paid, never downgrade to free

- LTV for a freemium customer:

$$F = Pr(\text{renew Free})(a + \delta F)$$

Recurring cash flow
from free

Freemium and LTV

41

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$$F = Pr(\text{renew Free})(a + \delta F) + Pr(\text{Paid upgrade})(p + \delta P)$$

Potential acquired cash
flows from paid

Freemium and LTV

41

□ freemium model:

□ F = free version, with ad revenues $\$a$

□ P = paid version, with price $\$p$

} Usually $a < p$

□ Assume: after upgrading to paid, never downgrade to free

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where

Potential acquired cash
flows from paid

$$\begin{aligned} P &= p \text{Prob}(\text{renew paid}) + \delta p [\text{Prob}(\text{renew paid})]^2 + \dots \\ &= p \frac{\text{Pr}(\text{renew Paid})}{1 - \delta \text{Pr}(\text{renew Paid})} \end{aligned}$$

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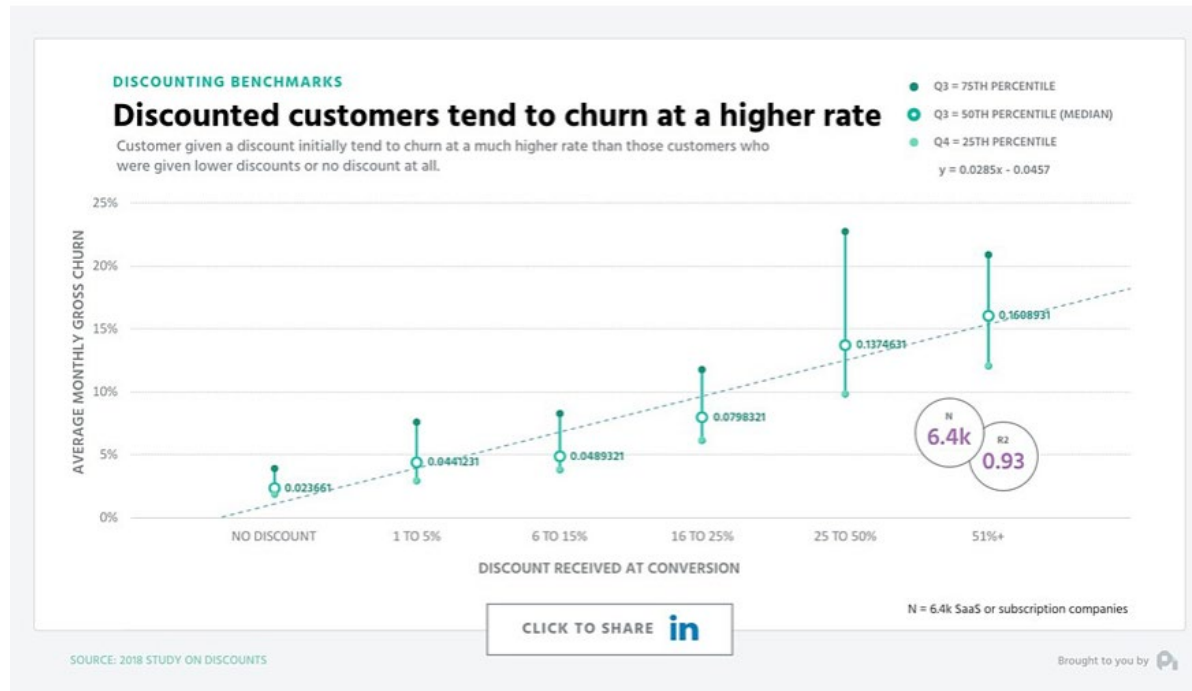
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Careful: $\text{Pr}(\text{renew Paid})$ probably depends on p and falls as $p \uparrow$

Selection effects of introductory pricing

Discounting to *juice* acquisition rates

42



Customers acquired via discount may be most likely to churn.
Makes sense, discounts mechanically *select* low-value customers at the margin!

➤ Optimized LTV pricing must balance **acquisition & retention**.

LTV at Ziprecruiter using simulated data.

- R code
- Results
- Figures

Data and code overview

44

- The actual Ziprecruiter data are confidential and cannot be circulated
- So in this Appendix I will provide R code you can use:
 1. to simulate hypothetical purchase data
 2. to estimate demand
 3. to compute optimal monthly prices
 4. to compute optimal prices against the LTV of the customer
- Although these are simulated data, I set them up to mimic the actual choice behavior from our experiment

The Logistic model

45

- To predict membership demand at different price levels, I estimated a “Logit” model
- The model predicts the probability of buying at a given price

$$\text{Prob}(\text{Pay for month of service} | \text{Price}) = \frac{\exp(\beta_0 + \beta_1 \text{Price})}{1 + \exp(\beta_0 + \beta_1 \text{Price})}$$

The Logistic model and NPV

46

- Suppose we have just run the experiment. We obviously do not know the repeat rates (i.e. retention rates) yet.
- If you need to make an immediate pricing decision, you will need to make an assumption about repeat behavior.
- For this exercise, let's assume trial and repeat rates are identical each month. For Ziprecruiter, we later verified this assumption after 6 months had elapsed (in fact, repeat rates were much higher than initial trial rates!)
- Now we can compute the NPV of revenues:

$$NPV(p) = \frac{1}{1 - \delta Prob(buy|p)} p Prob(buy|p)$$

A simulation in R to illustrate

Code to create simulated choice data

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- ❑ The next few slides replicate the logit simulation from the end of the elasticity slides in week 2

- `N = 8000` # number respondents
- `betatrue = matrix(c(-0.64,-0.004))` # inter. & price coeff
- `testprices=matrix(c(19,39,59,79,99,159,199,249,299,399))`
- `Price = sample(testprices,size=N,replace=T)` # randomize prices
- `epsilon = -log(-log(runif(N))) + log(-log(runif(N)))` # draw logistic errors
- `U = betatrue[1]+betatrue[2]*Price+epsilon` # buying utility
- `Y = (U>0)*1` # choice data

A simulation in R to illustrate Demand Estimation

48

- `### Estimate Logistic demand using R's built-in function "glm"`
- `out = glm(Y~Price,family="binomial")`
- `betahat = out$coefficients`

A simulation in R to illustrate

Plot demand estimates

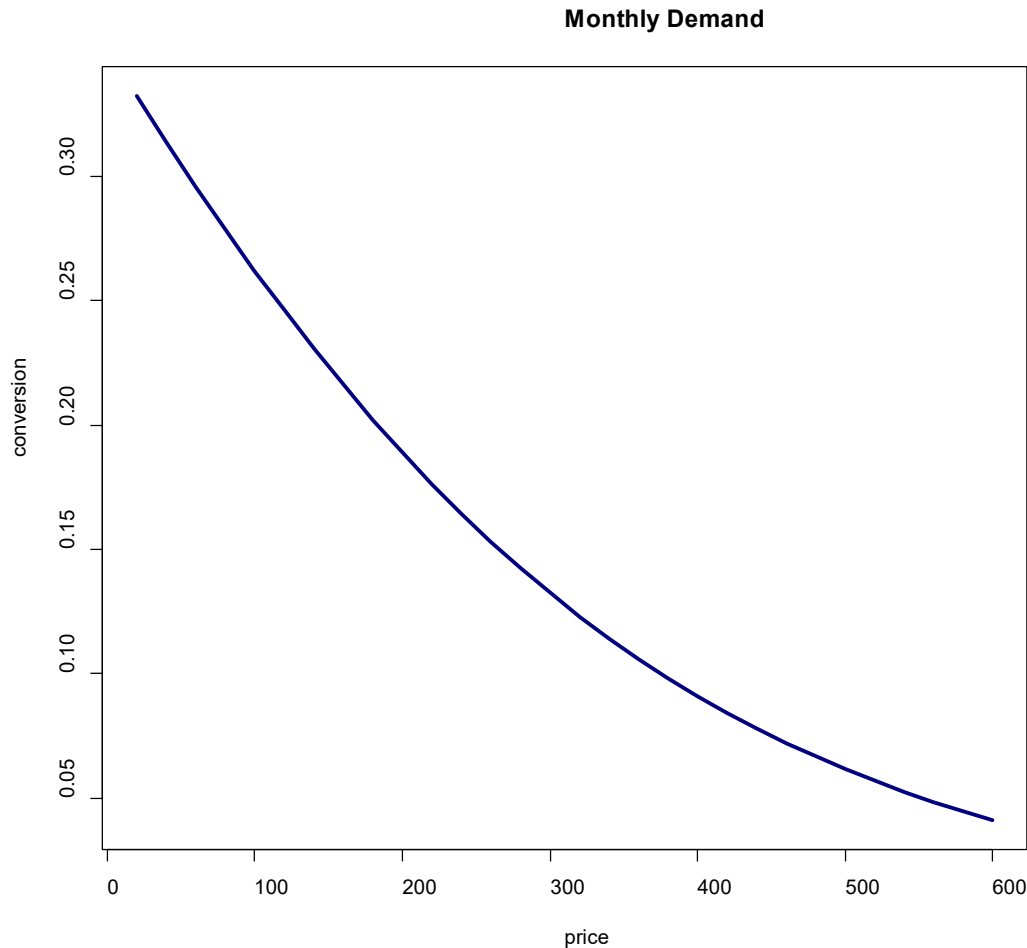
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- `# Function to compute logit probability`
- `share = function(x,beta){1/(1+exp(-beta[1]-beta[2]*x))}`
- `# Plot Predicted Demand along a price grid`
- `Pricegrid = seq(from=20,to=600,by=20)`
- `Qgrid = share(Pricegrid,betahat)`
- `plot(Pricegrid,Qgrid,main="Monthly
Demand",type="n",ylab="conversion",xlab="monthly price ($)")`
- `lines(Pricegrid,Qgrid)`

A simulation in R to illustrate

Plot demand estimates

50



A simulation in R to illustrate

Compute Monthly Revenue-maximizing price

51

- # Function to compute expected montly revenue
- `Erevenue = function(x,beta){share(x,beta)*x}`
- # Optimal static monthly price
- `Pstar = round(optimize(Erevenue,c(0,1000),beta=betahat,maximum=T)$maximum)`

A simulation in R to illustrate

Plot Predicted Monthly Revenue & optimal price

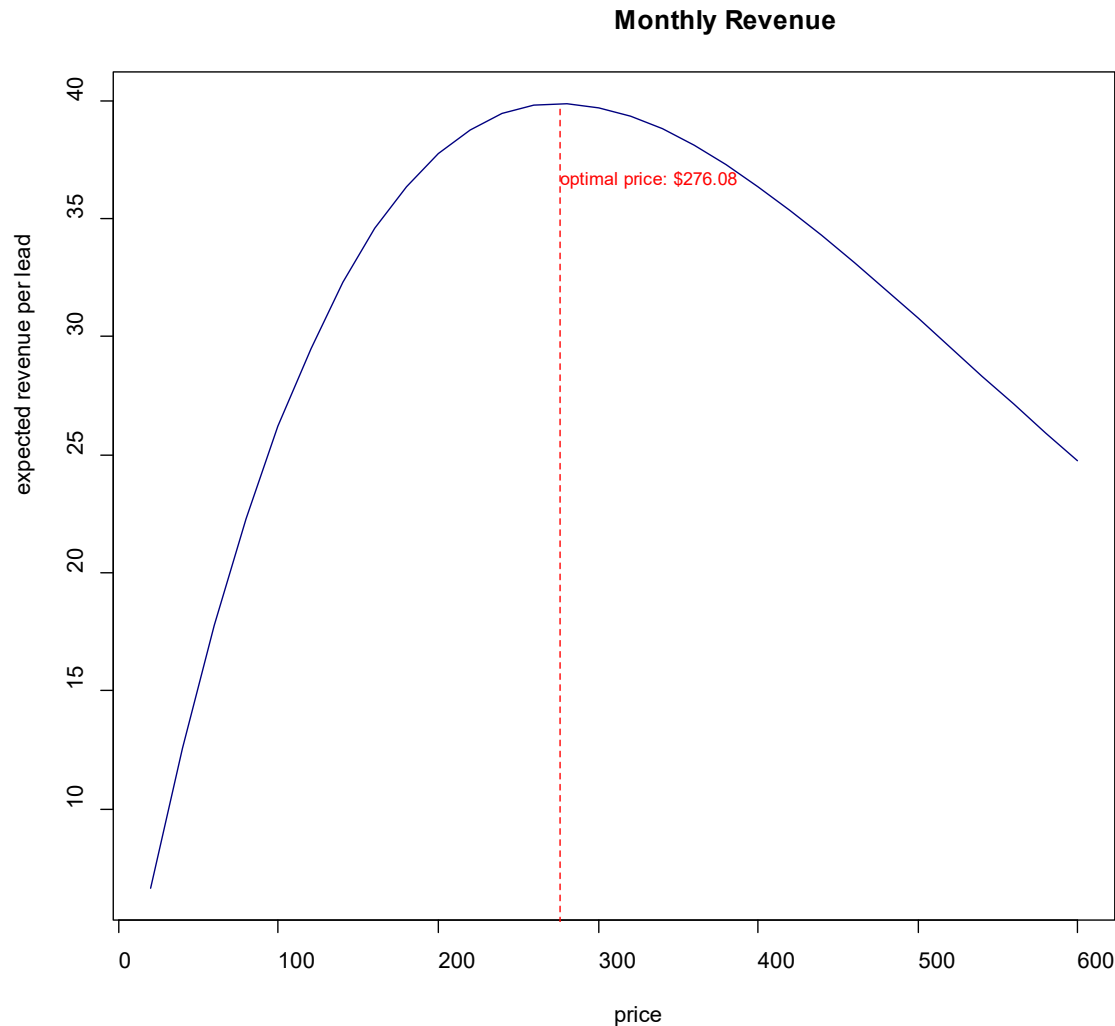
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- `#Plot Predicted Monthly Revenue`
- `Revgrid = Qgrid*Pricegrid`
- `plot(Pricegrid,Revgrid,main="Monthly Revenue",type="n")`
- `lines(Pricegrid,Revgrid,col="navy")`
- `lines(c(Pstar,Pstar),c(0,revenue(Pstar,betahat)),lty=2,col="navy")`
- `text(Pstar,revenue(Pstar,betahat),paste("optimal price:",
"$",Pstar,sep=""),pos=1,cex=.5,col="navy")`

A simulation in R to illustrate

Plot Predicted Monthly Revenue & optimal price

53



A simulation in R to illustrate

Compute NPV of Revenue maximizing price

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- #Function to compute NPV of Revenue
- $\text{NPVrevenue} = \text{function}(p, \text{beta}, \text{delta}) \{ 1 / (1 - \text{delta} * \text{share}(p, \text{betahat})) * (\text{share}(p, \text{betahat}) * p) \}$
- # Compute NPV-maximizing price
- $\text{delta} = .995$
- $\text{PstarLR} = \text{round}(\text{optimize}(\text{NPVrevenue}, c(0, 1000), \text{beta} = \text{betahat}, \text{delta} = \text{delta}, \text{maximum} = \text{T})\$maximum)$

A simulation in R to illustrate

Plot monthly revenue and NPV revenue functions

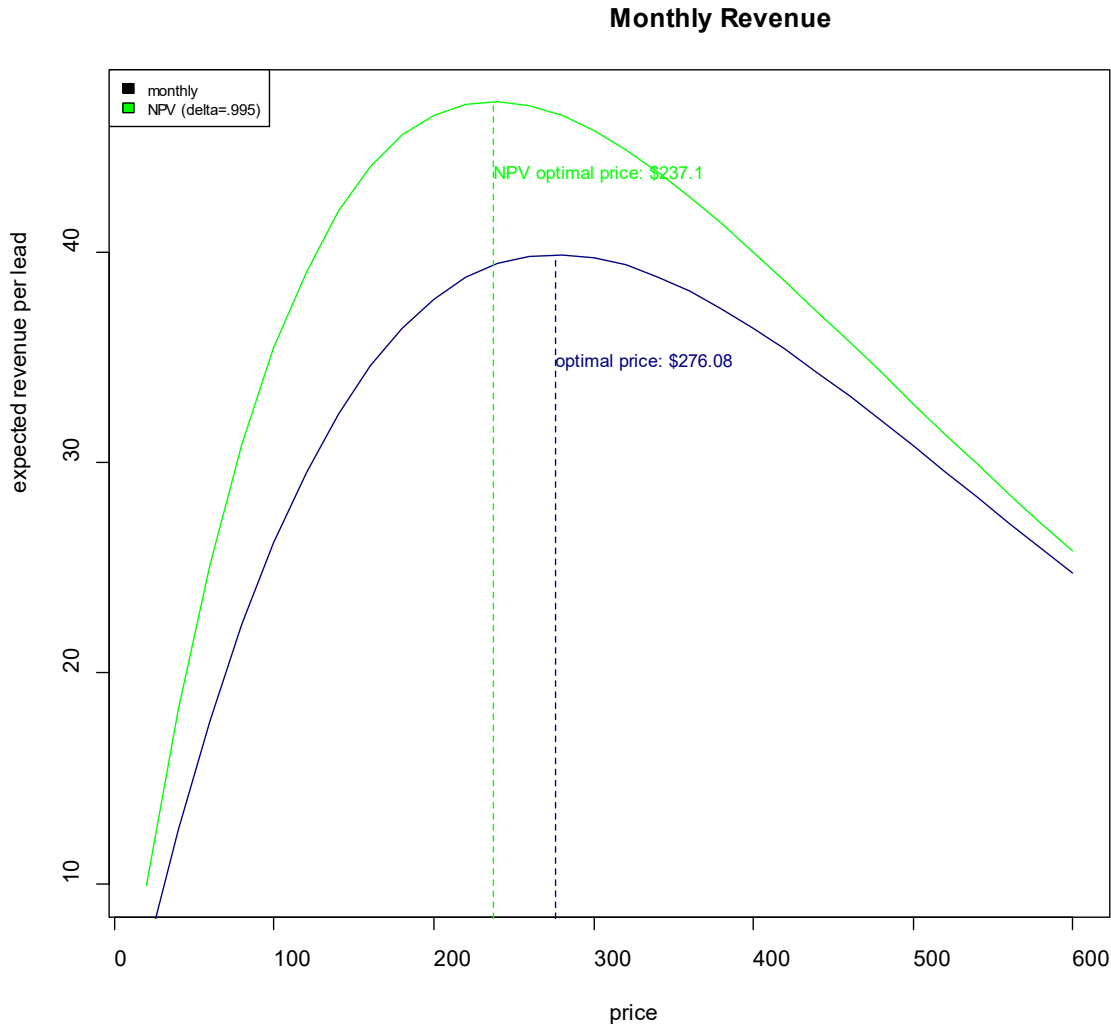
55

- `# Compute NPV of Revenues along the price grid`
- `NPVRevgrid = NULL`
- `for(tt in 1:length(Pricegrid)) {NPVRevgrid =
rbind(NPVRevgrid,NPVrevenue(Pricegrid[tt],betahat,delta))}`
- `#Plot Predicted Monthly Revenue & NPV of Revenue`
- `plot(Pricegrid,NPVRevgrid,main="Expected Monthly
Revenue",type="n",ylab="expected revenue ($)",xlab="monthly price ($)")`
- `lines(Pricegrid,Revgrid,col="navy")`
- `lines(Pricegrid,NPVRevgrid,col="green")`
- `lines(c(Pstar,Pstar),c(0,revenue(Pstar,betahat)),lty=2,col="navy")`
- `text(Pstar,revenue(Pstar,betahat),paste("optimal price:
$",Pstar,sep=""),pos=1,cex=.5,col="navy")`
- `lines(c(PstarLR,PstarLR),c(0,NPVrevenue(PstarLR,betahat,delta)),col="green",lty=2)`
- `text(PstarLR,NPVrevenue(PstarLR,betahat,delta),paste("NPV optimal price:
$",Pstar,sep=""),pos=1,cex=.5,col="green")`
- `legend("topleft",legend=c("static (delta=0)","NPV
(delta=.995)"),fill=c("black","green"),cex=.7)`

A simulation in R to illustrate

Plot monthly revenue and NPV revenue functions

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Extra

LTV at Ziprecruiter using actual retention rates

Pricing and LTV at Ziprecruiter

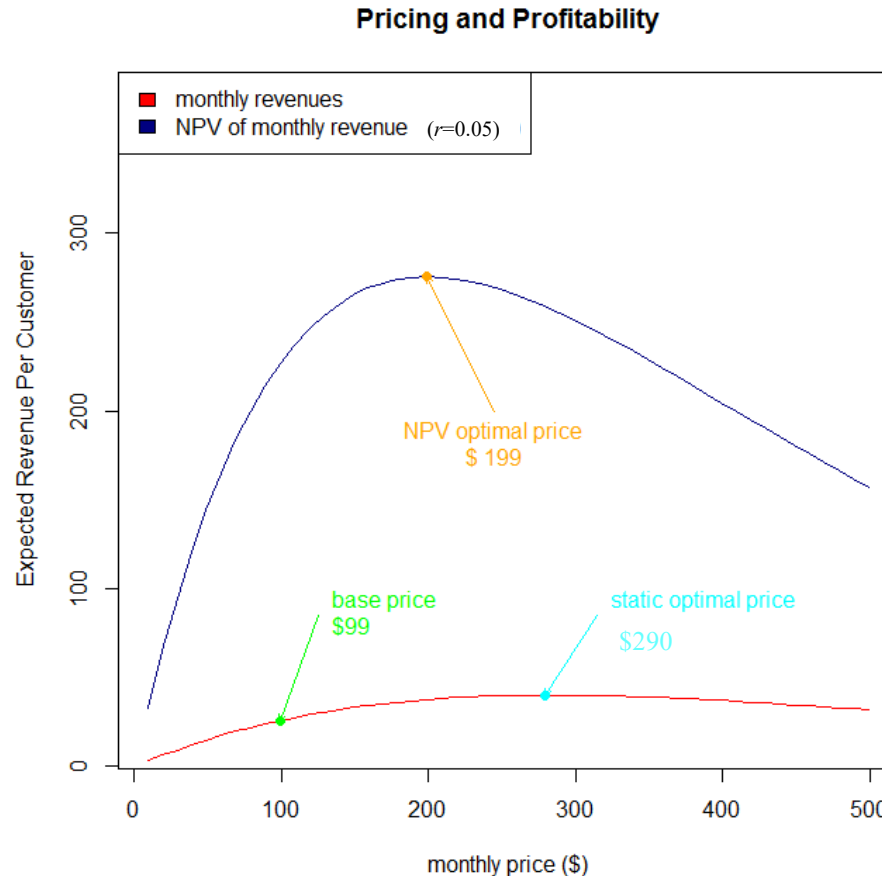
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- In practice, trial and repeat rates are probably different:
 - $Prob(trial|p) \neq Prob(retention|p)$
- 4 months after experiment, measure actual “retention” rate as function of price
 - ▣ At Ziprecruiter, retention rate higher than initial acquisition rate
 - ▣ Estimate repeat purchase probabilities for each of the first four months using the same customers from the experiment
 - ▣ assume repeat probability remains fixed after 4 months

Pricing and LTV at Ziprecruiter

use data to measure retention

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Optimal price for lifetime value even lower: \$199